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Boundary spanning between industry and university: the role of Technology Transfer Centres

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Abstract Technology Transfer Centres (TTCs) have been analyzed in the last few years by focusing on the relationship between a TTC, provider of knowledge-intensive services, and a firm client-receiver. Less attention has been devoted to a more complex relationship which involves in the dyadic provider-receiver tie a third relevant body, University. We provide both a theoretical and an empirical contribution by studying whether TTCs can bond the academic and industrial system and we define the activities that make-up this role such as: scanning and selection of R&D opportunities, bridge building, semantic translation of domain specific knowledge, co-production of new knowledge. The *boundary spanning role* of TTCs is discussed drawing on different and complementary theoretical perspectives. Moreover, we test research hypotheses on the antecedents of boundary spanning activity from a knowledge-based perspective. We argue that TTC boundary spanners need to leverage on both technical skills and networking competences. Empirical investigation has been carried out with a survey of the TTC population of North East Italy. The research findings highlight the task coordination activities implied by a boundary spanning role in joint R&D projects and show that the endowment of human capital at individual level and a qualified social capital at individual and organizational level are the main determinants.

Keywords Boundary spanning · Technology Transfer Centres · University-industry linkage · SMEs · Joint R&D projects · Human capital

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1 Introduction

The tendency of firms to implement an open innovation strategy is motivated by the benefits they attain in terms of innovativeness and reduced time-to-market (Enkel et al. 2009). Empirical research on this issue shows how multinational companies in traditional industries (Huston and Sakkab 2006) as well as firms in fast-moving-science and technology-based sectors (Chesbrough et al. 2006) deal with external sources of knowledge complementary to in-house R&D activities. Small and medium-sized enterprises (SMEs) operating in low-medium tech industries are also striving to improve their innovative outputs and escape the fierce competition based on the low costs of Asian competitors. They have matured an awareness of the potential advantages of opening the innovation process to external knowledge flows (Chesbrough and Crowther 2006; Enkel et al. 2009) and developing high tech products and processes with the help of universities. Research reveals that knowledge sharing among firms in different industries might improve innovative performance (Robertson and Patel 2007). In this regard, advances in technological fields such as biotechnology, nanotechnology, material sciences and alternative energy sources open new opportunities for cross-fertilization not only among high tech firms but also among low-medium tech SMEs (Yusuf 2008).

However, SMEs face higher barriers that hinder them from exploiting external sources, especially those that are not in the supply chain, such as research institutions. The exchange of scientific and technological knowledge between enterprises and universities is associated with higher uncertainties and information asymmetries that increase the search, bargaining and coordination costs (Kodama 2008). Moreover, SMEs attempting to recognize and assimilate external knowledge are confronted with obstacles such as resource scarcity, high cognitive distance, and low absorptive capacity (Muscio 2007; Kodama 2008).

Interactions with specialized local innovation intermediaries, namely Technology Transfer Centres (TTCs), may be of crucial importance for supporting innovative SMEs in establishing forms of collaboration with external organizations (Muscio 2007). TTCs encompass a set of diversified institutions, both public and private in nature, such as science parks, business incubators, research and testing laboratories, industrial liaison offices. Their mission is to be providers of knowledge-intensive services to firm-receivers in the different phases of their innovation process (Howells 2006), as well as to be part of a Technology Transfer (TT) infrastructure which promotes and facilitates networking activities in an industrial context. Therefore, they are usually embedded in the regional economy and their corporate governance tends to include stakeholders from the local public and private sectors (Barge-Gil and Modrego 2011).

Recent research in Europe has provided approaches (Reisman 2005; Howells 2006) and data that, as in the case of Italy (IPI—Istituto per la Promozione Industriale 2005), help to classify the different TTCs and their functions. A recent study has identified three main groups of services that these TTCs provide for their industrial clients (Spithoven et al. 2010). Firstly, they can be considered as a knowledge intelligence unit that helps firms articulate their needs, scan the environment and make connections between product and process innovation problems and new consistent solutions emerging across industries. This is the case of broker organizations like InnoCentive (Lakhani 2008) or Technology Transfer Offices (Muscio 2010). Furthermore, they can act as a knowledge agency performing R&D

activities for the receiver-firms. In this regard, Muller and Zenker in their study on Knowledge Intensive Business Service (KIBS) firms and their impact on the innovativeness of SMEs describe this knowledge re-engineering or recombination activity as follows "KIBS firms in fact act more and more as bridges and converters between technological and business expertise and localised knowledge and capabilities, becoming problem-solving actors specialised in the provision of the complementary knowledge inputs allowing the generation of innovations" (2001: 1505). The third function fulfilled by these Centres is that of knowledge repository, namely the collection, storage and diffusion of new technologies and knowledge developed in a specific industrial or geographical context, and information dissemination about new knowledge produced by firms together with its protection (IPRs) and its exploitation (commercialization).

Despite the increasing interest in the role of TTCs within the innovation system, the literature is still fragmented and further research is needed to fully understand their different functions. The literature concentrates principally on direct linkages between TTCs and firm-receivers, analysing the factors that enable firms to rely on these specialized organizations (Tether and Tajar 2008; Spithoven et al. 2010; Barge-Gil and Modrego 2011). Although this dyadic process is of great interest, there is, however, an issue that has only recently started to be investigated: the establishment of indirect industry-university links via TTCs acting as intermediaries (Yusuf 2008; Kodama 2008; Tether and Tajar 2008; Kirkels and Duysters 2010).

This specific bridging activity has been analyzed at a theoretical level by the literature on technology transfer intermediaries, but little empirical evidence has been provided until now (Chiaroni et al. 2008). Further, the large body of literature on industry-university relationships has devoted limited attention to the role that TTCs can play both in establishing industry-university links and in coordinating them.

As pointed out by Tether and Tajar (2008: 1093) "studies that look only at direct industry-academic links are missing an important part of the picture". In line with these authors, we argue that there is an unexplored activity of TTCs that can be of great value for industrial receivers, namely the partnership a centre could build between universities and firms through the joint management of innovative projects. SMEs, especially if they operate in low-medium tech sectors, seem less prone and able to collaborate on a project-base with academic partners. Thus, the bridging role of TTCs is particularly significant, since they facilitate the transfer of scientific and technological knowledge among partners that are otherwise distant and unrelated. However, due to the complex nature of this interaction, the intermediation role undertaken by a TTC between the academic and the industrial systems cannot be considered simply in terms of brokerage activities. Indeed, this role comprises several functions among those categorised in the literature on innovation intermediation (Bessant and Rush 1995; Hagardon and Sutton 1997; McEvily and Zaheer 1999; Howells 2006): scanning and selection of R&D opportunities, bridge-building, semantic translation of domain-specific knowledge, co-production of new knowledge. Thus, we maintain that TTCs that collaborate with universities potentially assume the important role of *boundary spanning* connecting two systems that do not easily come into contact with each other (Aldrich and Herker 1977; Tushman and Scanlan 1981; Carlile 2004).

The advantages and the complexity of the *boundary spanning* activity have been studied in the literature on innovation management by considering the individual roles of people or the networks between them (Aldrich and Herker 1977; Tushman and Scanlan 1981; Carlile 2004; Fleming and Waguespack 2007). More recently, the analysis has been extended to the organizational level in the networks between firms (Hagardon and Sutton 1997; Zaheer and Bell 2005; McEvily and Marcus 2005), and in industrial districts (Boschma and Ter

Wal 2007; Morrison 2008), but the role of *boundary spanning* in the technological transfer system is not yet fully understood.

Based on this gap in the literature, this paper addresses the following research questions: do TTCs that operate in contact with firms perform an activity of *boundary spanning* with universities? What are the antecedents of the *boundary spanning* of TTCs? In addressing these questions we extend the literature on technology transfer through a preliminary exploratory analysis of the characteristics and advantages of the *boundary spanning* role undertaken by TTCs, drawing on two different but potentially complementary perspectives; on one hand, the literature on inter-organizational technology transfer and, on the other, the literature on *boundary spanning*, traditionally focused on intra-organizational knowledge transfer. Furthermore, by adopting a knowledge-based perspective, we offer insights into the TTCs' endowment of resources and capabilities that enable them to act as a *boundary spanner* between industry and university.

The research hypotheses have been tested in a setting coherent with the research aims, namely North East Italy, which is made up of highly competitive regions from the point of view of innovation at the firm level, but which is composed principally of SMEs operating in low-medium tech sectors. In contrast with previous research, mainly based on in-depth qualitative studies or limited to only a few categories of TTCs, such as science parks, business incubators and academic technology transfer offices, which have been the most studied to date (Löfsten and Lindelöf 2002; Debackere and Veugelers 2005; Rothaermel and Thursby 2005; Muscio 2010), we carried out a survey on a population of all categories of TTCs, operating in the regions under investigation.

The paper is organised as follows. Section 2 begins with a conceptual discussion about the *boundary spanning* role of TTCs, then research hypotheses on antecedents are formulated. In Sect. 3 we present the research methodology and the measures used in the analysis. Section 4 offers some qualitative cases of TTC *boundary spanning*, reports the descriptive analysis of the sample and tests the hypotheses. Lastly, in Sect. 5 the paper concludes with a discussion of the findings, policy implications and suggestions for future lines of research.

2 Theoretical background

2.1 TTCs spanning the boundary between industry and university

The importance of external linkages for the innovation process has been highlighted in the literature and in empirical research, but firms aiming to exploit external ideas complementary to knowledge developed in-house, still encounter some problems, especially if they want to draw on sources that are not in the supply chain, such suppliers or customers, but are instead public research organizations, such as universities. Besides the importance of developing a network among the three systems for innovation (the triple helix university–industry–government relations) (Leydesdorff 2000; Dosi et al. 2006), particularly between universities and firms (Debackere and Veugelers 2005; Link and Siegel 2005; Decter et al. 2007), various studies maintain that these relations are well below their potentialities (Siegel et al. 2003; Yusuf 2008).

The difficulty of creating a market for technological knowledge (Arora et al. 2001; Dosi et al. 2006; Decter et al. 2007; Lichtenthaler and Ernst 2007) is one reason for this. In this regard, in a recent study on industry-university research collaboration in the Italian context, Abramo et al. (2011) found evidence for the presence of information asymmetry as a cause of inefficiencies. According to other studies, there are other reasons at an organizational level that can explain the low propensity to collaborate: the small size of a firm, its low absorptive capacity and the cognitive distance among partners (Laursen and Salter 2004; Fontana et al. 2006; Muscio 2007; Kodama 2008). Scarcity of resources and a short-term and an incremental orientation toward innovation, makes it difficult to work with universities on cutting-edge research projects which are characterized by high uncertainty and long-term orientation. Moreover, a low absorptive capacity implies a scarce ability to scan and recognize external sources of valuable knowledge. Finally, cognitive distance among firms and universities might negatively impact on any attempt to build a collaboration (Nooteboom et al. 2007).

The *boundary spanning* activity of TTCs with universities, is important, not only from the point of view of the individual centre but also from that of a regional innovation system. The relevance of boundary spanning activity can be understood by drawing on three different threads of research on innovation.

A first perspective is the research on clusters of innovation (Breschi and Malerba 2005). Some studies have shown the importance of the gatekeeper role in industrial districts, but they have focused on leading firms (Boschma and Ter Wal 2007). TTCs can also engage in *boundary spanning* among actors. On one hand, given their embeddedness in a specific geographical area, the Centres can foster and support technological spillovers and knowledge sharing by increasing the network density among the key actors of a region. On the other hand, as employers of qualified experts, some of whom are hired on a temporary contract basis, they can also enable the mobility of researchers within the area (Breschi et al. 2005). Therefore, a by-product the boundary spanning activity of TTCs is an increase in knowledge flows within a cluster.

Secondly, a perspective that brings the importance of the *boundary spanning* activity of TTCs to light is the social network analysis and, in particular, research on brokerage (Burt et al. 1998). According to these studies, a node that covers a *structural hole*, i.e., that activates a relation between two actors, who would otherwise not be in contact with each other, plays an important role since it allows the two partners to obtain knowledge which is not redundant compared to that already in their possession (Burt 2005; Zaheer and Bell 2005).

Thirdly, given the different nature of the knowledge to which TTCs have access through partnerships with universities, the *boundary spanning* role could imply not only the nurturing of information and knowledge flows within a region, and the activation of direct contacts among unrelated organizations (brokerage), but also the transformation of scientific knowledge into a language closer to the firms' communication codes (knowledge translation) (Carlile 2004). According to Tushman and Scalan (1981: 291) "communication across boundaries is difficult and prone to bias and distortion". Through the scanning of the market for emergent technologies and the engagement in joint projects with academic partners, TTCs nurture the firms' ability to assimilate and evaluate scientific knowledge and specifically the capacity to ascertain and assess science-based projects which are potentially complementary to their more applicative and short-term in-house projects.

In summary, TTCs can create the conditions for a better cognitive closeness between the two systems or facilitate the activation of interpersonal relations between firms and universities, two factors that appear to favour subsequent forms of face-to-face collaboration between them (Balconi and Laboranti 2006), as well as creating positive conditions for the coordination of multi-institutional projects (Corley et al. 2006; Fleming and Waguespack 2007).

Consistently with these arguments we suggest that the *boundary spanning* of TTCs implies two main processes: *information sharing*, namely the research, access and transfer of information useful for innovation (*informational boundary spanning*) (Tushman and Scanlan 1981) and *joint problem solving* or joint management of innovative projects (McEvily and Marcus 2005). The two processes are separate but not independent functions, since *information sharing* seems to be a requirement for *joint problem solving* (McEvily and Marcus 2005).

Thus, boundary spanning of TTCs could be "limited" to an activity of information exchange, or it can also comprise the joint management of innovative projects.

Informational boundary spanners "serve a dual function in information transmittal, acting as both filters and facilitators" (Aldrich and Herker 1997: 218). This role involves an investment in channels of communication and in environmental scouting abilities to identify more interesting and reliable sources. In the event of bridging between nodes whose knowledge distance is considerable, such as that between firms and universities, the function of information sharing could also require activities of translation, or rather decodification, of the information.

The second process of *joint problem solving* can create value both for TTCs and their receivers, but it requires greater interaction between the actor performing the role of *boundary spanning* and the nodes to which it is connected, since the knowledge is transformed (Carlile 2004) and consequently it is necessary to manage a greater relational interdependence and uncertainty. This type of process entails benefits for the Centre and its receivers not only in terms of key information gathered but of development of organizational capabilities (McEvily and Marcus 2005), status and influence through access to unique knowledge (Marrone et al. 2007).

On the other hand, it involves more coordination problems than information sharing activities. In a knowledge market for innovation, the complexity of coordination, that a *border-crossing* collaboration on a joint project entails for TTCs, is made more significant by variables that the organizational literature and the more recent literature on innovation and inter-organizational relations have highlighted.

A first variable is the interdependence between the actors, which, although it is already high in innovation processes, is even more problematic if managed by forms of collaboration between different organizations with distinct backgrounds and conflicting goals (Gulati and Singh 1998). A second variable derives from environmental uncertainty (volatility), as well as behavioural uncertainty (ambiguity) (Carson et al. 2006). The extent of uncertainty in a joint R&D project is higher than that implied by information sharing. Furthermore, recent studies have shown how, other than uncertainty, the complexity of the partitioning between actors and the interdependence of their actions increases due to the level of novelty of the technological knowledge, as in projects developed with universities (Carlile 2004). Lastly, a significant variable, particularly in inter-organizational relations, is constituted by the knowledge distance between the partners (Nooteboom et al. 2007), or rather the difference not only of technological knowledge but also of that between shared systems of perception, interpretation and evaluation, and of "shared meanings" linked to the organizational culture of each specific partner.

These considerations suggest that, for a joint project development to occur, it is necessary for TTCs to have the right competencies and resources in order to manage all the aforementioned problems. Building on these arguments, we explore the antecedents of the *boundary spanning* role for joint problem solving from a knowledge-based perspective.

2.2 Research hypotheses

In the literature on *boundary spanning* at an individual level of analysis the principal determinants are linked to individual attributes, such as personality (Burt et al. 1998) and technical competence (Tushman and Scanlan 1981). On the other hand, at an organizational level, a number of studies on collaboration between firms have explained the different propensity to technological sourcing and to joint problem solving, adopting a perspective linked to the wealth of organizational resources and capabilities, namely the characteristics of the firm's knowledge base and its human capital (Nooteboom et al. 2007; Brusoni et al. 2005).

We maintain that the resource endowment of TTCs can explain the propensity and the capacity of the Centres to undertake joint R&D projects with universities, hence to assume a *boundary spanning* role.

Boundary spanning activities are strongly connected to the qualification of the human capital in organizational units. Investment in graduates enables a TTC to benefit from the propensity of the internal personnel to establish social connections with the university staff thanks to their previous links and their cognitive closeness. In this regard, Borgatti and Cross claim that "research on homophily indicates that people are more likely to have social ties (especially strong ones) with those similar to themselves on socially important attributes such as… education" (2003: 434).

A second advantage in investing in internal qualified human capital is their greater ability to access knowledge with a high degree of abstraction and to manage projects at the technological frontier. According to Tether and Tajar (2008: 1082) "Science and engineering graduates are likely to be particularly valuable in accessing knowledge being developed in research organisations, and perhaps especially the public science-base which can be remote from the commercial pressures commonplace in firms".

Hence:

Hypothesis 1 The higher the TTC investment in internal human capital with high qualifications, the more likely it is to assume a *boundary spanning* role with a university.

The propensity to perform a *boundary spanning* role lies not only on the higher disciplinary competence of the internal personnel but also on the extent to which the TTC relies on external human capital.

First, external human resources are contractors (such as consultants or researchers) sought by a TTC to carry out project-related tasks that are not routine-based and require specialised capabilities not available among the TTC employees. Thus, they contribute to broaden the range of professional capabilities on which a TCC can draw for brokering activities. Second, ties between TTCs and external agents are weaker than those between TTCs and internal human resources. Therefore, external resources are more prone to engage in relationships across the boundaries of a TTC and to exchange with their own professional communities. Yet the professional cosmopolitism (Tushman and Scanlan 1981) of external agents may facilitate their *boundary spanning* behaviours. In summary, the wider range of specializations and the higher boundary spanning activity of a network of external contractors might positively affect the propensity of a TTC to screen and scout ideas developed elsewhere.

Consequently:

Hypothesis 2 The more a TTC relies on external human capital, the more likely it is to assume a *boundary spanning* role with a university.

Experience matured by a Centre in the activities of advanced R&D support constitutes an investment in a organization's knowledge base that reduces the cognitive distance with universities. The accumulation of prior knowledge of research activities allows the TTC to scan more quickly and reliably the external research and academic environment, to identify potential partners and mobilise the R&D capability in a mutual project with a university department or a group of researchers.

Furthermore, R&D investment could operate as a signal towards universities willing to find an intermediary for joint research projects with companies. These projects imply coordination problems not only due to task interdependence but also to ambiguity in the partner's behaviours (Carson et al. 2006). Management of such problems in a joint research project can be facilitated by conditions of trust (Perrone et al. 2003; McEvily and Marcus 2005). A university might have more trust in a TTC partner with a high reputation based on the quality or the type of research previously undertaken.

Therefore:

Hypothesis 3 If a TTC offers research activities among its services, it is more likely to assume a *boundary spanning* role with a university.

R&D capability might not be only one of the attributes of TTCs, thus a resource of their endowment, but it could also be a resource "available…as a function of their location in the structure of their social relations" (Adler and Kwon 2002: 18). First, a Centre which collaborates with other TTCs that invest in R&D could tap laboratories and expertise that they do not hold and that could be necessary for a joint project with universities and firms, thus it can have *access* benefits, according to studies on social capital (Burt 1992; Nahapiet and Ghoshal 1998). Indeed, the activation of external qualified relations makes it possible to discover information and knowledge which would not otherwise have been possessed. Second, a network of TTCs with R&D capabilities could also help in obtaining critical information more rapidly. Finally, we argue that collaborating with other TTCs that invest in R&D increases the identity and the status of the Centre within the innovation system, and, in turn, the likelihood of attracting and being chosen by a university which might select partners according to their position in the network. Recent empirical evidence offers support to this claim, finding that inventors who assume a *boundary spanning* role within an intra-organizational network are more likely to be selected by other inventors, since "an inventor with greater access to information about knowledge to be recombined is perceived as an important resource in knowledge-creation activities" (Nerkar and Paruchuri 2005: 774).

Hence:

Hypothesis 4 The more a TTC collaborates with other TTCs with R&D capabilities, the more likely it is to assume a *boundary spanning* role with a university.

3 Research methods

3.1 Research setting and data collection

The research is focused on the area made up of the regions in North East Italy (Veneto, Friuli Venezia Giulia and Trentino Alto Adige). The choice of these regions as research setting is due to the fact that they are important institutional actors in charge of drafting and implementing policies for technology transfer. Moreover, the choice of the three regions was motivated by the relative homogeneity of the industrial context, by their competitiveness at international level,[1] and by the widespread presence of small firms operating in low- medium tech sectors. In North East Italy microenterprises (1–9 employees) account for approximately 93% of the total population and the main manufacturing activities represented are the manufacture of basic metals, wood and paper, textiles (Istat 2010). Despite the small size of these companies and their specialization in traditional fields, they are striving to innovate their products and processes, and opportunities could be grasped by blending old technological fields with knowledge provided by research-based institutions, as in the case of nanotechnologies applied in metal or textile production.

As recent research into Italian industrial districts has shown, innovative families of technologies have been considered both attractive and feasible by low-medium tech firms. For instance, advanced polymers were considered useful for metal products and machinery districts, while fluoro-polymers for textile finishing and plasma treatment for natural fibres might be transferred to the silk production district (Roveda and Vecchiato 2008). Despite the need for these firms to access technologies which could potentially rejuvenate their industries, resource constraints, cognitive distance and low absorptive capacity might prevent them from establishing direct relationships with the academic providers of this new knowledge. In response to this need, over recent years, an increase in initiatives for intermediating firms and universities has been observed that distinguishes these three regions, even at a national level. Indeed, in comparison to large and high-tech firms which present fewer barriers to direct contact with academic partners, the fulfilment of a boundary spanning role by TTCs might be more effective in contexts such as those investigated in our research.

The field investigation was broken down into the following phases. Given the lack of a complete mapping of the TTCs operating in the regions considered, the first step of the empirical research was that of identifying the population to study by means of a desk analysis. The review of previous studies showed that the extant research concentrated on salient aspects but focused only on some actors (Balconi and Passannanti 2006; Compagno and Pittino 2006) or that it is, in many respects, dated compared to the dynamics in progress (Nest 2000). Therefore we consulted multiple sources[2] in order to identify all the potential Centres operating in the three regions. This preliminary mapping was integrated by internet sources (all the websites of the TTCs were analyzed) and structured interviews with the territorial actors through phone contacts, which allowed us to identify the Centres eligible to be included in the population. This first step, carried out between October 2006 and January 2007, led to the identification of 88 TTCs in the Veneto, 50 in Friuli Venezia Giulia and 10 in Trentino Alto Adige for a total of 148 TTCs. In the mapping phase we ascribed each Centre to one of the twelve typologies of TT institutions identified in the literature comparing the general mission of each category (see "Appendix") with the mission statement collected during the desk analysis and phone contacts. As reported in Table 1 in the geographical area under investigation roughly 58% of the TTC population is represented by laboratories supplying testing and analysis or R&D services. Moreover, in the three regions there are nine public research organizations, seven science parks, six

[1] According to the Regional Innovation Scoreboard (RIS), the three Italian regions show an innovation level at the EU average; in particular, they rank at high level in the innovation outputs indicators (Hollanders et al. 2009).

[2] IPI—Istituto per la Promozione Industriale 2005; Balconi and Passannanti 2006; NEST 2000; MIUR (Italian Ministry of University and Research); CNR (National Research Council); EBN (European Business Network); SINAL (Italian Authority for Laboratory Accreditation); SIT (Italian Calibration Service); SIL (Integrated System of Laboratories).

Table 1 Number and percentage of TTCs per typology (sample and population mapped)

	n	%	N	%
Laboratory—testing and analysis	16	24.62	52	35.14
Topic centre	10	15.38	16	10.81
Laboratory—applied research and technological development	9	13.85	34	22.97
Multi-sector centre	6	9.23	7	4.73
Public research organization	6	9.23	9	6.08
Business innovation centre	4	6.15	6	4.05
Chamber of commerce special agency and laboratory	4	6.15	6	4.05
Science park and technology hub	3	4.62	7	4.73
Territorial development enterprise	2	3.08	2	1.35
Business incubator	2	3.08	6	4.05
Technology transfer office	2	3.08	2	1.35
Experimental station	1	1.54	1	0.68
	65	100	148	100

business incubators, six business innovation centres and two academic technology transfer offices. We also mapped seven multi-sector centres and sixteen topic centres. The latter are institutions specializing in specific sectors or technological fields. In the regions analysed, the topic centres are specialized in traditional sectors such as agriculture, textile, wood products, footwear as well as in medium–high technologies like electronics and nano-technologies. We mapped six Centres that belong to the "special agencies" category and related laboratories of the Chamber of Commerce; while there are two territorial development enterprises. Finally, the only experimental station identified in the area provides services for the glass industry.

The second step involved a survey. A structured questionnaire was mailed to each key informant previously identified on the basis of their ability to respond to questions by virtue of her responsibility (chief executive officers, owners, technical or administrative directors of the Centre). The questionnaire was organized in three sections. The first section was aimed at collecting structural data (size, age, educational background of personnel, external partners, etc.); the second section focused on the main services provided and the actors served (types of firm, sector in which they operate), while the third section was devoted to collecting data on main joint projects with clients.

In order to maximize the response rate, an extensive follow-up was carried out. Mailing of the questionnaire was followed by a telephone call (at least 15 min for each of the Centres mapped) which illustrated the questionnaire and made the questions clear. This process was repeated at different time intervals to any remaining non-respondents. The response rate was equal to 43.92% for a total of 65 respondent TTCs (39 in the Veneto, 18 in Friuli and 8 in Trentino). The data collected from the questionnaire were integrated with the analysis of the web sites, documents obtained from the Centres as well as telephone calls that followed receipt of any questionnaire with incomplete answers (roughly 1 h for each of the 65 respondents). The variety of sources used permitted a process of data triangulation (Sonali and Corley 2006).

Table 1 shows in absolute and percentage values the number of TTCs per category as defined in the mapping phase. The distribution of the sample and the population are similar, suggesting that the sample is representative of the population of mapped Centres and, therefore, of the variety of typologies included in the definition of TTC.

We addressed the potential for non-response bias to provide assurance that the sample does not differ from the remainder of the population by comparing certain key attributes (geographical region and age) of respondents to those of the group of non-respondents. We accepted the hypothesis of independency between the variable “geographical region” and the variable “respondents” (p-value $= 0.051$). The low significance of the p-value can be ascribed to the difference between the sample and the group of non-respondents in the number of TTCs located in Trentino, the region with the smallest number of Centres but with the highest response rate. Welch Two Sample t test revealed no significant differences between the mean age (p-value $= 0.935$) of respondents and non-respondents. To further confirm the representativeness of our sample, we conducted a Kolmogorov–Smirnov Two Sample test. For the variable “age” we found no significant differences between the distribution of the sample and those of the non-respondents (p-value $= 0.984$).

The demographic analysis of the population depicts a stratified evolution with a proliferation of actors during the last two decades: the first Centres were set up in the 1950s (with the exception of a very small number of institutions which date back to an earlier date), but the system bloomed in the 1990s (27% of the Centres) and one-fifth of the population were established after 2000. The recent nature of TTCs in North-East Italy can be ascribed to the national and regional initiatives aimed at promoting an infrastructure to support the innovation process of SMEs and the networking between firms and public research organizations (Laranja 2009). This TT system differs considerably, for instance, from that in the US, where there is a long history of direct collaboration between science-based institutions and industry. In this regard, the two academic technology transfer offices analyzed were established only after 2000 as in the majority of Italian universities (Muscio 2010).

3.2 Operational measures

This section describes the variables used in our model.

3.2.1 Boundary spanning role

The survey asked the Centres to indicate the universities they have engaged in joint projects. We coded this information into a dichotomous variable that takes the value 1 if the TTC has indicated at least one university, otherwise the variable takes the value 0.

3.2.2 Internal qualified human capital

The variable is expressed using a rating varying from 0 to 1 which is obtained by calculating the average ratio of graduates over the total number of internal employees in the 3-year period 2004–2006.

3.2.3 External human capital

The external researchers variable is calculated as the average number of qualified people employed with atypical contracts over the 3-year period 2004–2006.

3.2.4 Research activity

The variable is dichotomous; 1 means that the TTC offers R&D services and 0 otherwise.

3.2.5 R&D Partners

Starting from the list of TTC partners with which the Centre collaborates on projects we checked if each partner has R&D services within its offering. The variable was calculated as the number of TTC partners with R&D capabilities.

3.2.6 Control variables

We also included in the model the age and the size of the Centres and an index which captures the relevance of medium–high sectors served by the TTC as control variables. The propensity to assume a *boundary spanning* role can be influenced by the age of the Centre (number of years since start-up). Young TTCs might have greater propensity to span boundaries, since they are less anchored by links with other organizations.

Moreover, recent research has identified size as a factor that positively influences firms' propensity to collaborate with universities (Laursen and Salter 2004; Fontana et al. 2006; Tether and Tajar 2008). Larger TTCs tend to have greater resources and more visibility within the innovation system, factors that can explain a higher propensity to assume a *boundary spanning* role with science-based organizations like universities. The size has been operationalized as the average number of internal employees during the 3-year period 2004–2006.

Finally, we controlled for a possible industry effect that can explain a higher propensity to span boundaries between firms and universities. If a Centre supplies a high percentage of firms operating in high and high-medium tech sectors, one might expect it to be more willing to collaborate with universities to satisfy the science-based requirements of these companies. However, high-tech clients might not require the intermediation services since they are able to articulate their own technological demands and to engage the academic partners in joint projects. We asked each Centre to indicate the industrial sectors supplied according to the NACE codes. To identify the sectors defined as medium–high and high tech (MHT) the OECD classification for manufacturing activities by type of technology was used (OECD 2003) as well as studies on science-based services (Marsili and Verspagen 2002). Therefore, we identified the following MHT sectors out of the total amount of sectors served by TTCs: DG Industries for the manufacture of chemical products and synthetic and artificial fibres; DK Industries for the manufacture of machines and mechanical equipment; DL Industries for the manufacture of electrical, electronic and optical machines; DM Industries for the manufacture of transport vehicles; telecommunications; IT services and research activities. This variable is expressed by an index that ranges from 0 to 1, where 0 means that a Centre supplies only medium–low and low tech sectors (MLT) and 1 means that a Centre provides services only to MHT sectors.

4 Results

4.1 TTC boundary spanner between university and industry

A first result concerns the emergence of TTCs' role of intermediation between the university system and industry.

All the Centres in the sample are specialized providers of technological knowledge for SMEs operating in North East Italy. The client-firm orientation of the Centres, expressed by the supply of advanced services, advice and other different forms of assistance for business activities, is supported by the mission statement defined during the mapping phase, and the high percentage weight of the turnover for services supplied over the total sources of finance (on average 62.8% of the total income). The services offered range from technological research to design and prototypes, from technological diagnosis to brokerage, from incubation to trials and laboratory testing. As far as the industrial features of the client-firms are concerned, only 3 out of 65 Centres provide TT services exclusively to high and medium–high tech firms, 20 only to low and low-medium tech, and the remaining 42 TTCs to both technological classes. Almost the entire client portfolio is made up of micro-small enterprises (up to 50 employees) mainly located within the province or within the regional boundaries. TTCs also offer their services to a limited number of medium-large enterprises which are leading players in their sector. However, R&D projects which involve large companies entail the engagement of a network of small companies which act as suppliers and can benefit from the knowledge the project aims to generate.

With regards to their relationship with academic partners, 70.8% of the TTCs assume the *boundary spanning* role collaborating for joint projects with at least one university (the majority of the Centres, 55.4%, has indicated up to two universities). Table 2 shows in absolute and percentage values the number of Centres per number of universities with which they have developed R&D projects.

Due to their institutional role in the regional system and the presence of academic actors as stakeholders in the corporate governance of the Centres (such as science parks, technology transfer offices, business incubators or public research organizations), there might be typologies of TTCs more willing to establish collaboration linkages with the academic system. However, we found that Centres belonging to all the twelve categories mapped (see Appendix and Table 1) fulfil a boundary spanning role even though at different levels. For instance, 31.3% of the laboratories for testing and analysis engaged in joint projects with universities; this percentage is higher in the case of multi-sector centres (50.0%), territorial development enterprises (50.0%) laboratories for applied research (66.7%), and Chamber of Commerce special agencies and laboratories (75.0%).

The qualitative data collected through the survey (we asked the Centres to provide information on their most significant R&D projects during the 3 year period 2004–2006), integrated with telephone contacts and secondary information from the websites of the TTCs, enabled us to deepen our understanding of their bridging activity.

Table 2 Number and percentage of TTCs per number of universities with which they collaborate

Number of universities	Development of joint projects	
	n TTCs	%
0	19	29.23
1	26	40.00
2	10	15.38
3	6	9.23
4	3	4.62
>4	1	1.54
	65	100

We report some cases of Centres which act as boundary spanners by managing projects that involve other TTCs and universities with subsequent application of the research outcome to the industrial system or projects with both academic and industrial partners. The Centres offer TT services to local SMEs operating in typical regional sectors such as the eyewear, textile, ceramic and construction industries.

Alpha is the national institute for optical product accreditation, its areas of activities cover training, accreditation, regulation and applied research services. It is a multi-sector centre since it supplies services to several industries (e.g., optical and eyewear products, jewellery manufacturers, eye and face protection). The three main projects described by Alpha lasted 3–4 years and were financed by regional and European funds as well as by the firm-partners. Among these projects, one aimed at applying physical vapor deposition and plasma-enhanced chemical vapor deposition techniques in the eyewear product. Alpha coordinated the academic institution involved in the project, another four Italian TTCs and two firms operating in the nanotechnology field. Another project concerned the implementation of the Laser Enhanced Plating (LEP) technique, aimed at obtaining metal deposits without incisions in eyewear products, jewellery and costume jewellery, thus preventing counterfeiting. A global leader which produces machines for goldsmiths and laser machines for welding joined the project. The academic organizations that took part in the project are two polytechnic universities.

The following statement from an academic partner of Alpha highlights their effective interactions, and clarifies the "coordinator task" played by this centre between university and industry:

> We have continued contacts with Alpha, a continuous osmosis of ideas and sharing of tools. At Alpha they are very good at coordinating research of different types, particularly in the field of those projects directed both regionally and inter-regionally, even in the midst of many difficulties. The centre attempts to create occasions, opportunities and partnerships. Alpha goes from door to door in search of partners, and convinces them (given that there is a little reticence in entering into projects), either because it is the firms' first experience in this field or because they are too small and they are afraid of not being good enough. Many projects envisage a technological transfer, if there are firms that participate less actively in the partnership, they still benefit from the effects of research.

Beta is a topic centre which offers research and testing services to the firms belonging to the supply chain of the textile and clothing sectors. One of the projects in its portfolio involved, in collaboration with an academic partner and other Italian TTCs, a research project lasting 16 months which aimed to apply the techniques instrumental to the use of natural colorants, studying their environmental compatibility and evaluating the effects of the tested colors on health.

Gamma is a laboratory that provides testing services for the ceramic and construction sectors. With the collaboration of two universities, one TTC and seven firms, it coordinated a 2-year project that aimed to find alternative raw materials and new processes in order to reduce the environmental impact in the ceramic industry.

The qualitative evidence helps clarify the *boundary spanning* role of TTCs. They promote projects that aim to advance cutting-edge technology and apply the research outcomes especially to SMEs in low-medium tech industries. Thus, the firm-partners are helped to engage in processes of cross-fertilization across disciplines. Furthermore, they do not only activate contacts but also carry out task coordination activities managing the relationships among multiple stakeholders.

4.2 Descriptives

Table 3 displays the descriptive statistics and the bivariate correlation coefficients between the variables included in the model.

As discussed in the analysis of the setting, the majority of the Centres in the sample were established after the 1990s. The mean age of the TTCs is about 19 years. As reported in Table 3, older TTCs are more likely to be larger. The Centres have a flexible structure with an average of 18 employees, half of whom are graduates. The correlation matrix shows that older TTCs are more likely to invest in qualified human capital. The qualitative data we collected in some TTC boundary spanners shows that the majority of graduates hold a degree in engineering or in economics. They are in charge of the different activities carried out by the Centres, such as innovation diffusion, incubation, training, specialized services (such as certification, industrial design, new materials, prototyping). Personnel employed in testing, analysis, rapid and virtual prototyping laboratories are usually technicians who do not necessarily hold a degree. The interviewees highlighted the importance of both technical and managerial competences when the Centres are engaged in projects with firms and universities. The technical skills enable the internal employees to give precise indications about the technological demands of the firms during the audit sessions and to match it with what the academics can supply. As for managerial competences, project management is necessary to guarantee an efficient and effective accomplishment of the aims of the projects, while relational skills are important in building reciprocal trust and fostering mutual understanding among multiple stakeholders.

Where the external human capital is concerned, the descriptive statistics show a high variability among the Centres. Indeed, although the mean is 10, the median value is 1–2 collaborators and 75% of TTCs rely on less than four collaborators. From the interviews it turned out that external agents are mainly consultants with significant experience in the business context, self-employed professionals specialized in specific disciplines (technologists in mechanics and material science or industrial designers), and former R&D managers of local leading companies who are currently retired. These external agents provide the Centres not only with the access to knowledge not held by the internal personnel but also with professional capabilities in dealing with the entrepreneurial and academic communities. The following statement of one of the boundary spanner Centres interviewed highlights the role fulfilled by external human capital:

> Their contribution is to pinpoint the technological problem of the company, to work it out and to discuss it with the academic departments in order to acquire the appropriate competences.

The descriptive statistics report that 58% of the Centres have matured experience in providing R&D services to firms, and the networking with partners with R&D capabilities is limited on average to 1–2 partners. The descriptives also show a positive and significant correlation between these two variables. Moreover, Centres with a higher number of external contractors are more likely to provide R&D services and collaborate with other TTCs with R&D capabilities. Furthermore, the presence of R&D services and the networking with TTCs with R&D capabilities turned out to be positively correlated with the size of the Centres.

In order to examine multicollinearity, we calculated the variance inflation factor (VIF). VIFs are between 3.25 and 1.11, which is below the rule-of-thumb cutoff of 5. The VIF mean is just above 1, thus issues of multicollinearity do not seem to prompt concern.

Table 3 Descriptive statistics and correlation matrix

	Mean	Std. dev.	Min.	Max.	1	2	3	4	5	6	7	VIF[a]
1. Boundary spanning	0.71	0.46	0	1								
2. Internal qualified human capital	0.53	0.27	0	1	0.354***							1.16
3. External human capital	10.04	30.31	0	175	0.423***	0.227						1.40
4. Research activities	0.58	0.50	0	1	0.419***	0.194	0.250**					1.33
5. Partners for joint projects (n = 64)	1.61	1.99	0	10	0.362***	0.067	0.385***	0.381***				1.32
6. Age	19.32	20.48	1	133	−0.103	−0.245**	0.131	0.034	−0.028			2.76
7. Size	18.07	36.61	0	262	0.108	−0.029	0.116	0.354***	0.275**	0.458***		3.25
8. MHT sectors	0.45	0.35	0	1	−0.095	0.174	−0.062	−0.108	−0.079	−0.141	−0.110	1.11

n = 65; ** $p < .05$; *** $p < .01$

[a] VIF (Variance Inflation Factor) is computed regressing each variable on all the other explanatory variables, calculating the R2 and then VIF = 1/(1-R2). High values of VIF correspond to high multicollinearity and are of concern when the VIF is 5 or higher

4.3 Hypotheses testing

The focus of this section is the estimation of the determinants of the *boundary spanning* role of TTCs.

The regression analysis was carried out in two steps. Firstly, we estimated the model using all the observations reduced to 64, due to the presence of a missing value in correspondence to the variable "R&D Partners". The analysis of the residuals of the model and Cook's distance suggested that we consider one observation as an outlier. The standardized residual of this observation is very large (−7,750) and its Cook's distance is 0.946, thus it can be considered as influential and potentially might have an effect on the estimation of the coefficients. Our qualitative data suggests that this TTC, due to its institutional role of regional promoter of the innovation activities in the industrial system, significantly differs from the other centres in terms of resource endowment.

In the second step, we re-estimated the model excluding the outlier The results are summarized in Table 4, which presents the estimated values of the parameters and the relative *p*-values obtained using Gretl software.

The overall analysis shows the positive and significant effect of the independent variables, other than Research Activities (Hypothesis 3), on the probability a TTC assumes a *boundary spanning* role. In Hypothesis 1 we predicted that a higher investment in internal qualified human capital through the recruitment of graduates increases the likelihood that a TTC would play a *boundary spanning* role with a university. The parameter of the model for this independent variable is positive and the *p*-value is 0.077. This result should be considered with some caution.

We proposed in Hypothesis 2 that the higher the reliance on external human capital, the more likely a TTC would be to span boundaries with a university. The results confirm this hypothesis (*p*-value = 0.04). Thus, as predicted, by collaborating with a wide external network of researchers TTCs increase their likelihood of playing a *boundary spanning* role with universities.

Contrary to expectations, in the model the coefficient for the Research Activities was not statistically significant. We predicted a positive effect of this antecedent on TTC *boundary spanning* role. The overall results and our qualitative data might provide a

Table 4 Results of regression analysis

	Coeff. (S.E)	*p*-value
Const	−1.883 (1.005)	0.061*
Internal HC	2.945 (1.664)	0.077*
External HC	0.806 (0.393)	0.040**
Research Activities	0.295 (0.852)	0.729
R&D Partners for JP	0.678 (0.336)	0.044**
Control variables		
Age	0.009 (0.027)	0.747
Size	0.003 (0.051)	0.960
MHT industries	−1.343 (1.168)	0.250
Number of cases	63	
Chi-square (*p*-value)	30.8692 (0.0001)	
	50	
Correctly predicted (accuracy rate)	(79.4%)	

* Significant at 10%;
** significant at 5%;
*** significant at 1%

possible explanation for the lack of support found for Hypothesis 3. TTCs might not need to be providers of R&D services to bridge the industrial and the academic system since they could draw the technical capabilities a project requires from university partners and other TTCs. A further investigation, instead of considering merely the presence of R&D activities within the range of services offered by the Centres, could take into consideration the number of hours devoted by staff to R&D activities.

In Hypothesis 4, we examined the relationship between the number of partners of TTCs with R&D capabilities and the propensity of a Centre to collaborate with academic institutions. The coefficient turned out to be positive and significant (p-value $= 0.04$), giving support to this hypothesis.

Finally, examining the regression coefficients for control variables, the age and the size of the Centres were not significant. Moreover, also the relevance of clients of TTCs operating in medium–high tech industries was not significant. This result confirms that they do not have a propensity to support specifically those companies that operate in science-based sectors.

5 Conclusions

This research has been geared towards providing a first investigative contribution about the *boundary spanning* role of TTCs, as yet little studied in the literature, but potentially significant for the purposes of SMEs innovation and policies concerning local innovation systems.

This work contributes on two levels. Firstly, the paper enriches the literature on technology transfer by adopting different, though potentially complementary, perspectives and studies: on the one hand, the literature on technology transfer and inter-organizational collaborations for innovation and, on the other, the literature on *boundary spanning*, which traditionally considers individual roles and focuses on the problem of knowledge transfer at intra-organizational level. Furthermore, the paper contributes to broadening the empirical base of the studies on technology transfer and innovation intermediaries by means of the investigation of a sample of TTCs which encompasses a higher number of operating categories.

A second contribution is given by the empirical, though still exploratory, results on the *boundary spanning* role of TTCs and its antecedents.

The descriptive and qualitative analysis have highlighted the existence of a *boundary spanning* role in the sample of TTCs in North East Italy. The significant aspect that emerges from the results is the high percentage of the sample that fulfils the complex, though with greater added value, activities of joint management of innovation projects with universities and industrial partners. Only a few studies have focused their attention on the definition of the roles or functions undertaken by these intermediaries within the TT infrastructure (Howells 2006; Spithoven et al. 2010). From our empirical analysis we observe that, independently of the categories the Centres belong to (i.e., research laboratories, science parks, business innovation centres), they carried out the bridging activity between the academic and industrial systems. As recent research has suggested, there is a need in the technology transfer and innovation literature to look beyond the traditional dyadic and linear provider-receiver relationship (Chiaroni et al. 2008; Laranja 2009). In order to investigate the complex interactive learning process of technology transfer in which TTCs can intervene not only as brokers or producers of knowledge but also as boundary spanners, further studies should delve into the processes of *boundary spanning* by a qualitative approach.

The analysis of the determinants from a knowledge base perspective allows us to draw several considerations. *Boundary spanning* appears to be an activity conducted by TTCs with an endowment of human and social capital. The results seem to highlight the specificity of these organizations as regards human capital and the relevance of qualification of internal personnel. Findings show that the presence of graduates might facilitate the Centres' access to university knowledge due to the cognitive closeness of TTC staff with academic researchers, and their ability to translate the scientific language and to transform the abstract knowledge into applicative projects.

These explorative results, however, are based on the measurement of the investment in qualified human capital as the percentage of graduates over total internal employees. This could limit our understanding of the specific technical skills and the relational competences personnel require to manage boundary spanning activities, such as evaluating the potential of academic and industrial partners and dealing with negotiation as well as coordination problems. Future research should investigate this issue and consider in-depth the qualification of the internal staff that support TTCs in their activities, not only in terms of competences in technical disciplines but also in terms of culture, cognitive frameworks and project management capacities which are useful in joint projects between universities and firms.

TTCs that span boundaries between firms and universities are those that not only invest in qualified internal personnel but also in external collaborators. The investment in a network of qualified experts increases the capability of a TTC to gather information and to scout technology. Professional cosmopolitism of these external collaborators may facilitate their *boundary spanning* behaviours, thus they could contribute to the propensity of the TTC to establish linkages with the academic system for joint projects. A more complete picture of the effect of external human capital on TTC *boundary spanning* requires further fieldwork in order to understand the characteristics of the researchers employed with atypical contracts, and the role they play in the joint projects. Furthermore, a relevant issue is the network of relationships linking TTCs, external experts and universities. Future research should investigate which kinds of contract and relationship connect a TTC and an outside agent and the features of that network of ties with other contractors and institutions.

The results highlight the impact of networking at an organizational level. Indeed, when a TTC invests in its social capital via networking with other Centres with R&D capabilities, it increases its probability of accessing qualified information through them, and it improves its status and reputation within the innovation system. This in turn impacts on the probability of collaborating with universities and on the capacity to manage multi-institutional projects.

In conclusion, this study draws attention to the multiple roles a TTC can undertake within the local innovation system, focusing in particular on the still neglected activity of stimulating linkages between SMEs and universities for joint projects. Particularly in geographical contexts which lack longstanding and close collaborations between the industrial and the academic systems, boundary spanning Centres take on a crucial role, which is beneficial to the local economy. Policy-makers should be aware of TTCs and their activities, identify those Centres which can fulfil the *boundary spanning* role and support initiatives aiming to increase their effectiveness, by providing incentives to TTCs and enhancing their human capital and social capabilities which enable them to orchestrate numerous interfaces, cross-disciplinary teams and stakeholders.

The research has some limitations that should be areas of future research. Firstly, the meagre number of TTCs in the sample, despite the response rate of 43.92%, and the geographical characterisation of the setting. Although replication of the findings in a larger sample would be welcome, we wish to point out that TTCs might be different in other regions or countries in terms of size, corporate governance and other characteristics, but the roles they fulfil in the system of innovation seem to be the same, especially in Southern European countries (Spithoven et al. 2010; Laranja 2009). Secondly, by considering only the resource endowment of the TTCs we overshadowed the impact of the structural positioning of the Centres within the innovation system. Future research can benefit from social network analysis. In light of the most recent studies on inter-organizational collaborations (Zaheer and Bell 2005), the role of corporate capabilities and the positioning in the network as complementary determinants of the role of *boundary spanning* could be studied together. Lastly, the high number of TTC boundary spanners in the sample raises the issue of potential overlapping and inefficiencies in local systems. Given the exploratory nature of our research, we did not investigate the conditions under which TTC boundary spanners fulfil their role effectively. Future studies should provide insights on the determinants of an effective *boundary spanning* role.

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Appendix

See Table 5.

Table 5 Typologies of TTCs mapped and mission statement

Typologies	Mission statement
Experimental station	Carrying out pre-competitive industrial research and development analysis, testing and experimentation of products, processes and new technologies, technical and scientific dissemination of knowledge and documentation in specific sectors at the national level (IPI—Istituto per la Promozione Industriale 2005).
Science park and technology hub	Promoting the economic development and competitiveness of regions and cities by creating new business opportunities and adding value to mature companies; fostering entrepreneurship and incubating new innovative companies; generating knowledge-based jobs; building attractive spaces for the emerging knowledge workers; enhancing the synergy between universities and companies (International Association of Science Parks).
Technology transfer office	Supporting the academic staff to identify and manage the organization's intellectual assets, including protecting intellectual property and transferring or licensing rights to other parties to enhance prospects for further development (European Commission 2004).

Table 5 continued

Typologies	Mission statement
Business incubator	Accelerating the growth and success of entrepreneurial companies through an array of business support resources and services that could include physical space, capital, coaching, common services, and networking connections (National Business Incubation Association).
Business innovation centre	Offering a range of integrated guidance and support services for projects carried out by innovative SMEs, thereby contributing to regional and local development (European Business and Innovation Centre Network).
Chamber of commerce special agency and laboratory	Furthering the development and expansion of technological innovation through the offer of services that meet the requirements of the firms associated with the Chamber of Commerce (IPI—Istituto per la Promozione Industriale 2005).
Territorial development enterprise	Gathering and coordinating scientific, organizational and financial resources in the region in order to transfer acquired information into new production processes and research results to the entrepreneurial context.
Topic centre	Promoting a specific industry or a specific technological area inside a geographical context (IPI—Istituto per la Promozione Industriale 2005).
Multi-sector centre	Supplying diversified services to firms operating in several sectors (IPI—Istituto per la Promozione Industriale 2005).
Public research organization	Performing research in its own Institutes, promoting innovation and competitiveness of the national industrial system, promoting the internationalization of the national research system, providing technologies and solutions to emerging public and private needs, offering advice to Government and other public bodies, and contributing to the qualification of human resources (Italian National Research Council).
Laboratory	Supplying qualified services of research and development, analysis and testing to client firms.

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